

FACULTY OF SCIENCE

MSc in Particle Physics and Cosmology

Polarization of the CMB: Deciphering E- & B-Modes

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1 Introduction

The Cosmic Microwave Background Radiation (CMBR) is the oldest light in the Universe, emitted during the epoch of recombination at $z_* \simeq 1100$ (approximately 380,000 years after the Big Bang). The CMBR is a nearly perfect blackbody radiation that fills the entire sky, with a temperature of about 2.725 K. The "nearly" is a key word as the CMBR presents some small deviations or distortions generated from the primordial fluctuations in the early Universe. These perturbations are expanded by the inflation epoch and generate the Large Scale Structure (LSS) of the Universe.

The most well known distortions are the temperature anisotropies, which are scalar perturbations of the order of μK and have been measured with great precision. But these are not the only anisotropies, the CMBR also presents polarization quadrupole anisotropies that are generated not only by the scalar (compressional) perturbations, but also by vector (vorticity) and tensor perturbations (gravitational waves). They are of the order of several μK and are generated by the Thomson scattering of the CMB photons with the free electrons in the presence of quadrupole anisotropies. The polarization of the CMBR can be decomposed into two components: E-modes and B-modes, which will be explained in detail in the next sections.

Then, it is crucial to note that the CMBR has been altered during its journey through the Universe. This implies that the CMB photons that we receive today are a distorted version of the original blackbody radiation emitted at recombination, due to the mentioned scattering, among other processes such as gravitational lensing, foreground contamination, etc.

A more sophisticated way to extract information from the CMB photons is through the power spectrum, which is the Fourier transform of the two-point correlation function of the CMB anisotropies. There are many different power spectra that can be computed from the CMB anisotropies, such as the temperature power spectrum, the E-mode power spectrum, the B-mode power spectrum, and the cross-correlation power spectrum between temperature and polarization. These will be discussed in detail in the next sections.

Finally, from all these measurements we can extract information about the growth and evolution of the Universe, such as the inflationary epoch and the values of the cosmological parameters. These can be in agreement with the Λ CDM model, the most widely accepted model of cosmology, or they can present some tensions that could indicate new physics beyond the standard model of cosmology.

2 Physics of CMB Polarization

The CMB photons are the same kind of photons that we receive from stars, but they have a much lower energy and a much longer wavelength due to the redshift of the Universe. These can be scattered as any other photon, and the more likely event due to their low energy is an elastic scattering with free electrons, which is known as the Thomson scattering process, a special case of Compton scattering in the low-energy limit.

The light that travels towards us, which I will define as the z -direction, oscillates in the plane perpendicular to the direction of propagation, the $x - y$ plane. If the intensity of the light is not the same in both directions (there is a quadrupole anisotropy), then the scattered light will be polarized.

The polarization of the CMBR can be described by the Stokes parameters I , Q and U , which are defined by the matrix:

$$I_{ij} = \begin{pmatrix} I + Q & U \\ U & I - Q \end{pmatrix} \quad (1)$$

where I is the intensity of the light, and Q and U are the linear polarization parameters. Unpolarized light has $Q = U = 0$, while linearly polarized light has $Q \neq 0$ and/or $U \neq 0$.

These Stokes parameters are not invariant under rotations, so they depend on the choice of the coordinate system. To overcome this problem, we can decompose the polarization field into two components: electric (E-) and magnetic (B-) modes, which are invariant under rotations and have different parity properties.

It is customary to make the flat-sky approximation to describe the polarization field, which is valid for small angular scales. In this approximation, we can define the Fourier transform of the polarization field as:

$$P(\ell) = \int d^2\theta P(\theta) e^{i\ell \cdot \theta}, \quad (2)$$

where ℓ is the multipole moment, which is the Fourier conjugate of the angular position θ on the sky. Defining the azimuthal angle ϕ_ℓ of the 2D wavevector $\ell = (\ell_x, \ell_y) = \ell_0(\cos \phi_\ell, \sin \phi_\ell)$, the E-mode and B-mode become:

$$E(\ell) = \cos(2\phi_\ell)Q(\ell) + \sin(2\phi_\ell)U(\ell) \quad (3)$$

$$B(\ell) = -\sin(2\phi_\ell)Q(\ell) + \cos(2\phi_\ell)U(\ell) \quad (4)$$

On one hand, the E-modes are curl-free and have even parity, so they produce a scalar $E(\ell)$. These are generated by the scalar perturbations of the CMB and are the dominant source of polarization in the CMB. On the other hand, the B-modes are divergence-free and have odd parity, so they produce a pseudo-scalar $B(\ell)$. The latter are generated by the tensor perturbations and are about one order of magnitude weaker than the E-modes.

The polarization patterns of the E- and B-modes can be visualized in Fig. 1 for a single plane-wave with wavevector $\ell = \ell_0 \hat{\mathbf{e}}_x$ in the x -direction.

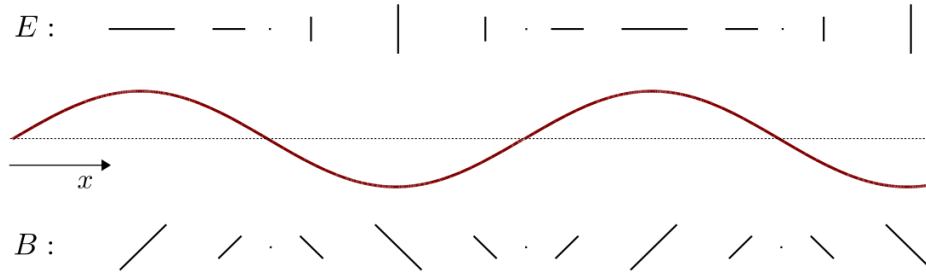


Fig. 1. Polarization generated by a single plane-wave perturbation along the x-axis for an E-mode (top) and a B-mode (bottom). The length of each headless arrow indicates the strength of polarization. Photons are propagating out of the page (along the z-axis). FIGURE 10.3 and footnote adapted from [1].

The E-mode amplitude changes parallel or perpendicular to its orientation. On the contrary, the B-mode varies in strength in a 45° angle from its orientation. Another more realistic way to visualize them is with a superposition of radial plane waves in the $x - y$ plane, which is shown in Fig. 2. The E- and B-modes properties mentioned above can also be observed in this figure.

The radiation field presents a tiny quadrupole before recombination because photons and electrons were tightly coupled, and only those photons that last scattered in an optically thin region could present a quadrupole anisotropy. As a result, the linear polarization of the CMB is smaller than the temperature anisotropies and depends on the duration of the last scattering. To visualize and better understand the process of polarization of light, Fig. 3 represents the process for different types of incoming radiation.

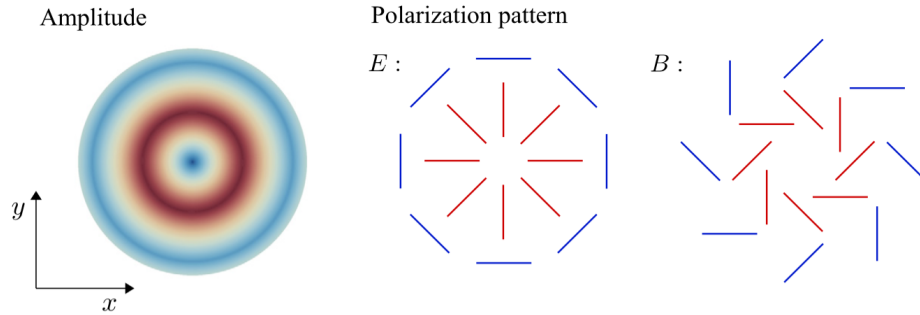


Fig. 2. Polarization generated by a superposition of radial plane waves in the $x - y$ plane, with equal phase and amplitude but different azimuthal angles ϕ_ℓ (left panel). The red lines represent the peaks and the blue lines the troughs of the radial wave. The middle panel shows the E-mode pattern, which is curl-free and has even parity, while the right panel shows the B-mode pattern, which is divergence-free and has odd parity. Photons are propagating out of the page (along the z -axis). FIGURE 10.4 and footnote adapted from [1].

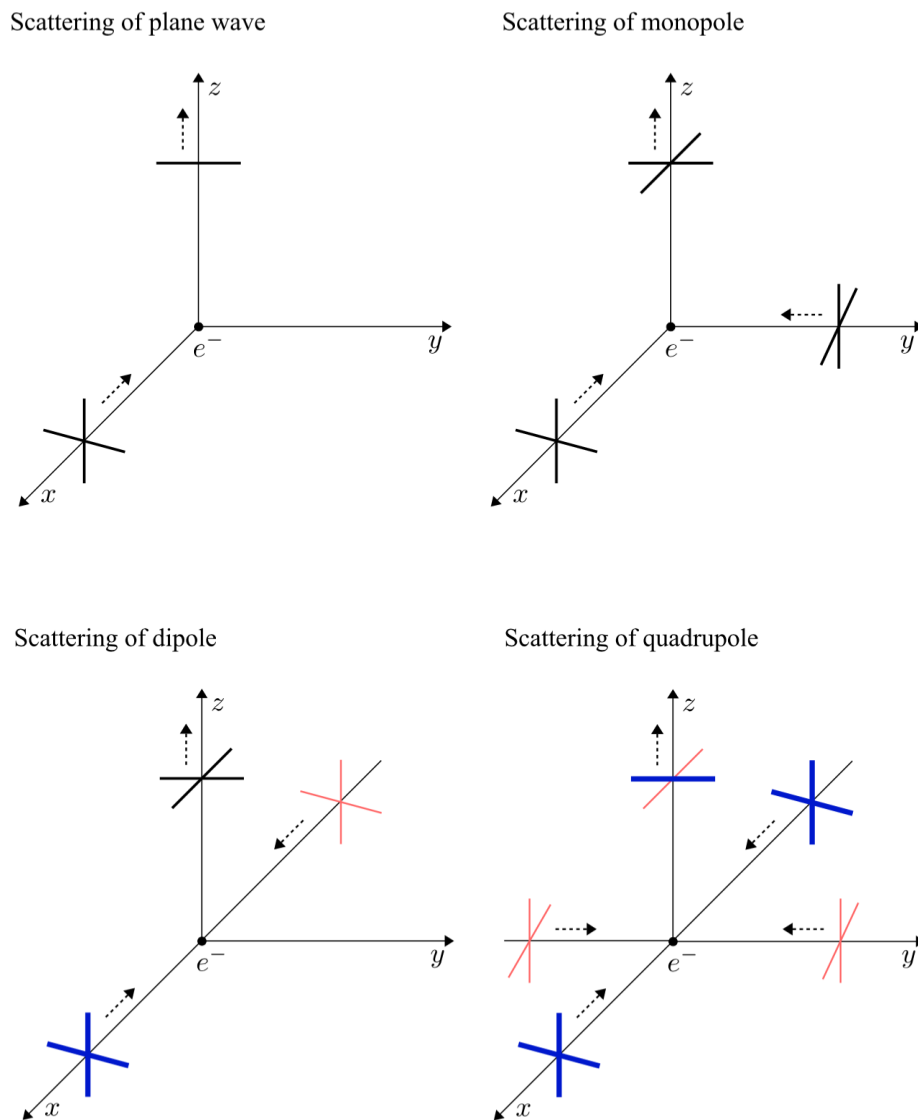


Fig. 3. Thomson scattering of photons with an unpolarized plane wave produce perpendicular polarization (top left panel). Isotropic (monopole) radiation produces no polarization if the incoming amplitude is uniform (top right panel). Dipole (bottom left panel) radiation produces unpolarized light if the incoming amplitude is uniform. Quadrupole (bottom right panel) radiation produces polarization as the incoming amplitude is not uniform. Heavy blue (red) represent hotter (colder) radiation.

It is always assumed that the incoming radiation (to the electrons) is unpolarized, so the outgoing radiation will be polarized if there is a quadrupole anisotropy in the incoming radiation.

In order to depict the different quadrupolar polarization patterns generated by the scalar, vector and tensor perturbations of the CMB, it is useful to use the spherical harmonics Y_l^m , as I will explain in the next subsections.

2.1 Scalar Perturbations

The scalar perturbations of the CMB are the energy density perturbations. They are the only fluctuations that can form the LSS through gravitational instability, and they are the dominant source of temperature anisotropies in the CMB. These are the main source of the temperature anisotropies and the E-mode polarization, as they have the biggest amplitude among the three types of perturbations. However, the scalar perturbations do not generate B-mode polarization because of their symmetry properties, as they are invariant under parity transformations (so they have even parity), while the B-modes have odd parity.

This type of perturbation generates quadrupole anisotropies with $(l, m) = (2, 0)$, which are represented by the following spherical harmonic:

$$Y_2^0(\theta, \phi) = \sqrt{\frac{5}{16\pi}}(3\cos^2\theta - 1), \quad (5)$$

and whose polarization pattern can be described as a "pure Q-field" with no U component,

$$Q = \sin^2\theta, \quad U = 0, \quad (6)$$

This pattern generates a polarization that is parallel or perpendicular to the z -axis, which corresponds to the E-mode polarization. It can be visualized in Fig. 8 of A.1.

2.2 Vector Perturbations

The vector perturbations of the CMB decay rapidly in the early Universe, so their contribution to the polarization is negligible compared to the scalar and tensor perturbations as their amplitude decreases faster.

This type of perturbation generates quadrupole anisotropies with $(l, m) = (2, 1)$, which is represented by the spherical harmonic

$$Y_2^{\pm 1}(\theta, \phi) = \sqrt{\frac{15}{8\pi}}\sin\theta\cos\theta e^{\pm i\phi}, \quad (7)$$

The polarization pattern generated by the vector perturbations is the following:

$$Q = -\sin\theta\cos\theta e^{i\phi}, \quad U = -i\sin\theta e^{i\phi}, \quad (8)$$

and can be visualized in Fig. 9 of A.2.

2.3 Tensor Perturbations

The tensor perturbations of the CMB are the primordial gravitational waves generated during the inflationary epoch. They can generate both E-mode and B-mode polarization, but they are expected to be much weaker than the scalar perturbations, so their contribution to the CMB polarization is also expected to be small. The detection of the B-mode polarization generated by the tensor perturbations would be a smoking gun for inflation, as it would provide direct evidence of the existence of primordial gravitational waves.

Quadrupole anisotropies with $(l, m) = (2, 2)$ are generated, which are represented by the following spherical harmonic:

$$Y_2^{\pm 2}(\theta, \phi) = \sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{\pm 2i\phi}, \quad (9)$$

The polarization pattern generated by the tensor perturbations is the following:

$$Q = (1 + \cos^2 \theta) e^{2i\phi}, \quad U = -2i \cos \theta e^{2i\phi}, \quad (10)$$

and can be visualized in Fig. 10 of A.3.

3 E-modes

In this section I will focus on the E-mode polarization, the dominant source of polarization in the CMB, and I will explain their origin and what information we can extract from them.

To pass from scalar, vector and tensor perturbations to the E- and B-modes, one must take into account the parity properties of the perturbations and the polarization patterns they generate. Following the reasoning and the explanation of the previous section, the E-modes can be generated by the scalar perturbations in the density (energy) in the early Universe (to say, the LSS), which create the quadrupole anisotropy that later polarizes the CMB photons linearly.

However, scalar perturbations are not the only ones that can generate E-modes; vector perturbations can generate E-modes with smaller amplitude than B-modes (although they are negligible as they decay rapidly in the early Universe), and tensor perturbations generate E- and B-modes with similar amplitude. The ratios are $B/E = 0, 6, 8/13$ respectively, as can be seen in Fig. 4.

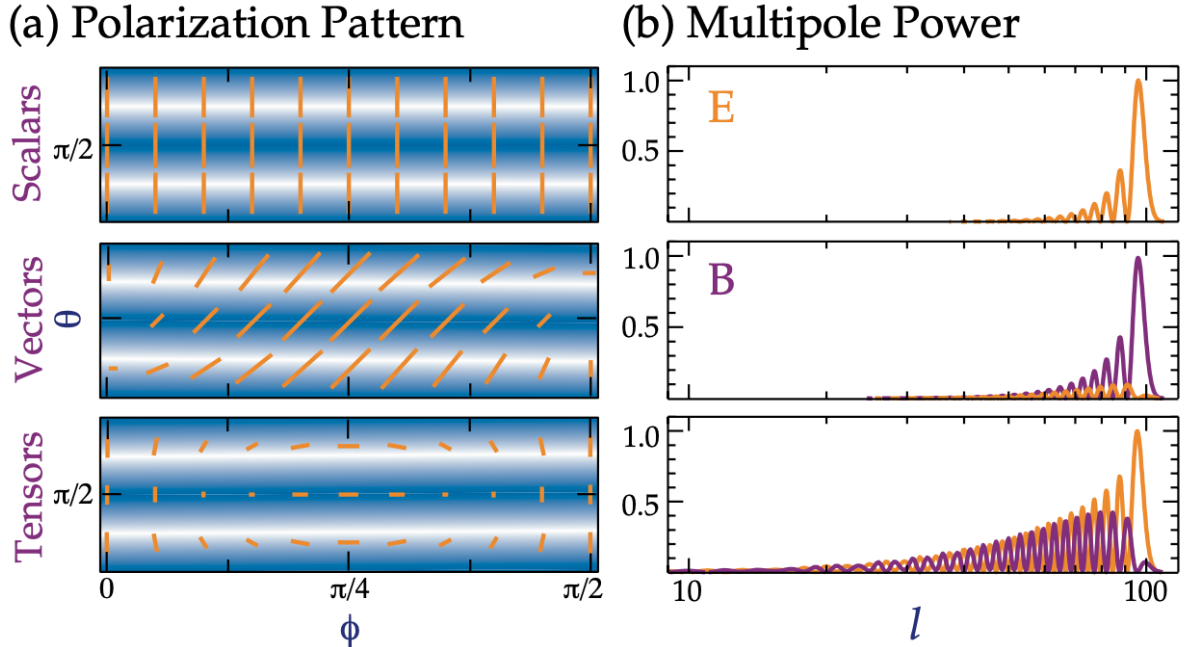


Fig. 4. The E and B components of a plane-wave perturbation. (a) Modulation of the local E-quadrupole pattern from scattering by a plane wave. As discussed before, modulation parallel or perpendicular to the polarization generates an E-mode with higher ℓ , while modulation in the crossed (45°) direction generates a B-mode with higher ℓ . (b) The scalar perturbation generates only E-modes, the vector perturbation generates B-modes mainly, and the tensor perturbation generates both E- and B-modes with similar amplitude. while the vector and tensor perturbations generate both E- and B-modes with different ratios. Fig. 11 and footnote adapted from [2].

To better understand the generation of E-modes from the scalar perturbations, Fig. 5 is shown.

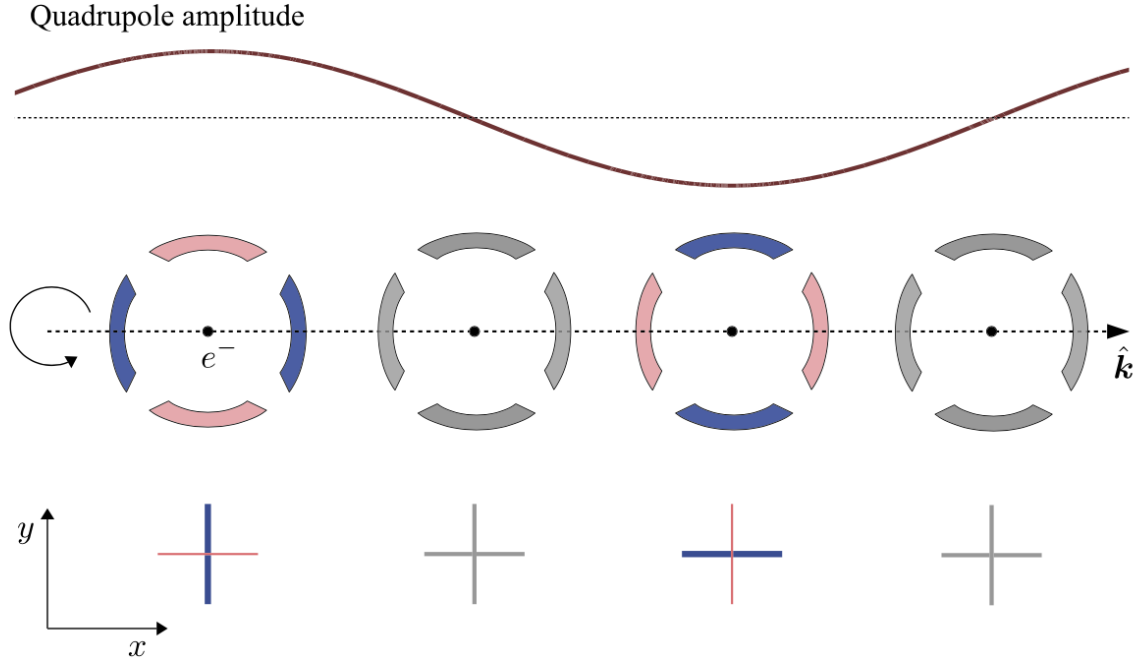


Fig. 5. Polarization pattern generated by a scalar perturbation in the x -direction (top panel), which generates a quadrupole anisotropy in the temperature pattern seen by the electrons (middle panel), which subsequently generates a linear E-mode polarization pattern (bottom panel) when observed in the z -direction (coming out of the page). FIG. 10.8 and footnote adapted from [1].

To deepen the understanding of the E-modes, it is useful to analyze the power spectrum of the E-modes, C_{EE} , which is the Fourier transform of the two-point correlation function of the E-mode polarization field. Before computing the power spectrum, we must deduce the polarization moments expression for the E-modes, $E_\ell(k)$, which is given by

$$E_\ell(k) \simeq -\frac{5k}{6\tau'(\eta_*)}\Theta_1(k, \eta_*)\frac{\ell^2}{(k\eta_0)^2}j_\ell(k\eta_0), \quad (11)$$

where Θ_1 is the dipole moment of the temperature anisotropies, k is the wavenumber of the perturbation, η_* is the conformal time at recombination, η_0 is the conformal time today, and j_ℓ is the spherical Bessel function of order ℓ . The deduction of this expression can be found in Chapter 10 of [1], which assumes a tight coupling between the photons and the electrons before recombination, and that the polarization amplitude is equal to the amplitude of the E-mode polarization: $\Theta_P = \Theta_E$.

However, there is a superposition of many plane waves with different amplitude, wavenumber, k , and direction, $\hat{n}$. As it is usually done for the temperature anisotropies, we define the transfer function as

$$\mathcal{T}_\ell^E(k) \equiv \frac{E_\ell(k)}{\mathcal{R}(\mathbf{k})} \quad (12)$$

where $\mathcal{R}(\mathbf{k})$ is the primordial curvature perturbation. Then, the power spectrum of the E-modes is computed as

$$C_{EE}(\ell) = \frac{2}{\pi} \int_0^\infty k^2 dk |\mathcal{T}_\ell^E(k)|^2 P_{\mathcal{R}}(k), \quad (13)$$

where $P_{\mathcal{R}}(k)$ is the power spectrum of the primordial curvature perturbation.

Temperature anisotropies are also generated by the scalar perturbations, then there is a correlation between the temperature anisotropies and the E-mode polarization. The temperature power spectrum is denoted as C_{TT} and is computed as

$$C_{TT}(\ell) = \frac{2}{\pi} \int_0^\infty k^2 dk |\mathcal{T}_\ell^T(k)|^2 P_{\mathcal{R}}(k), \quad (14)$$

where $\mathcal{T}_\ell^T(k)$ is the transfer function for the temperature anisotropies. The cross-correlation power spectrum between the temperature anisotropies and the E-mode polarization is computed with the following expression:

$$C_{TE}(\ell) = \frac{2}{\pi} \int_0^\infty k^2 dk |\mathcal{T}_\ell^{T*}(k) \mathcal{T}_\ell^E(k)| P_{\mathcal{R}}(k). \quad (15)$$

However, the standard convention to plot the cross-correlation power spectrum (and many other power spectra) is defined as

$$\mathcal{D}_\ell^{TE} \equiv \frac{\ell(\ell+1)}{2\pi} C_{TE}(\ell), \quad (16)$$

where the factor $\ell(\ell+1)/2\pi$ is introduced to make the power spectrum more flat and easier to visualize. All this power spectra are shown both in Fig. 6 and in the middle panel of Fig. 7 for the temperature-polarization cross-correlation (TE) case.

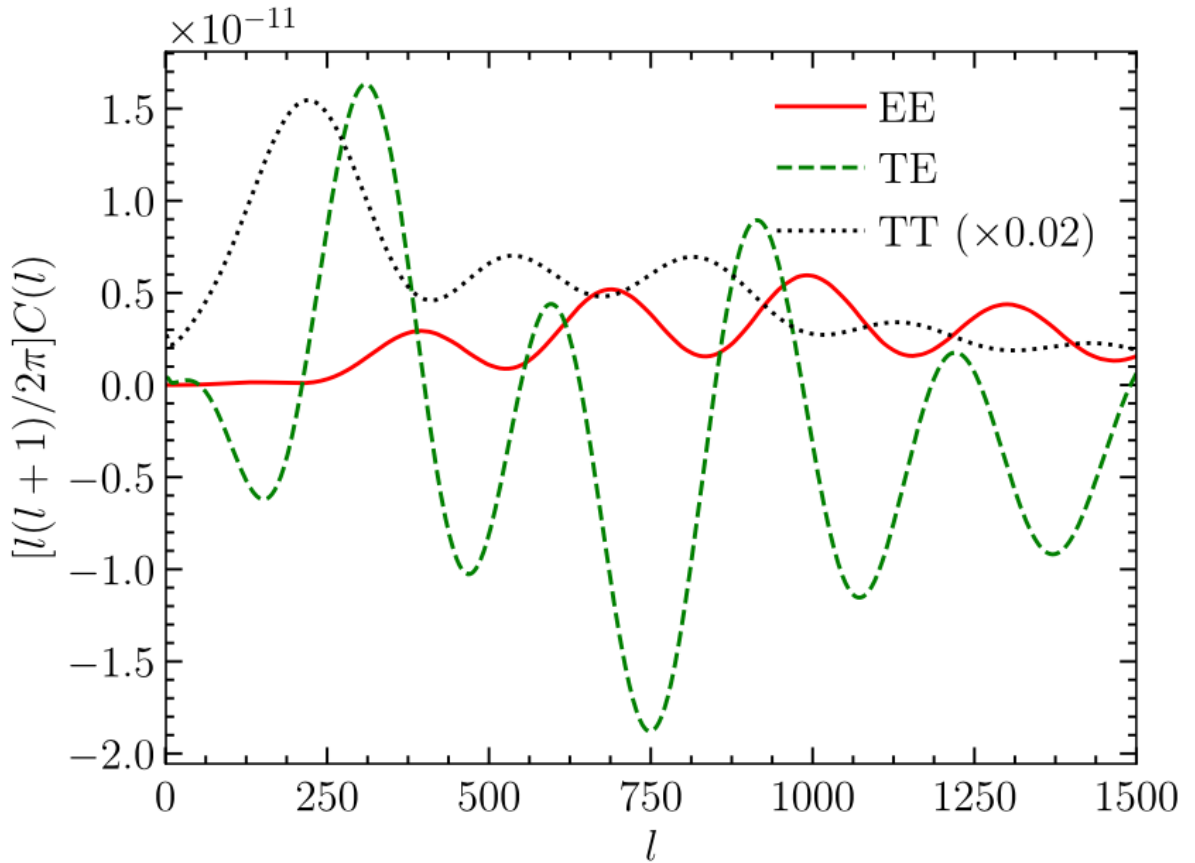


Fig. 6. Angular power spectra of temperature anisotropies (dotted black), E-mode polarization (continuous red), and temperature-(E-mode)polarization cross correlation (dashed green) for the fiducial concordance cosmology. The temperature power spectrum is about a factor of 1/50 larger than the E-mode one. FIG. 10.9 and footnote adapted from [1].

The acoustic oscillations on $C_{EE}(l)$ are caused by the dipole of the photon distribution, while the ones on $C_{TT}(l)$ are generated by a combination of monopole and dipole. As the monopole is dominant in the latter, the peaks of both power spectra are out of phase.

4 B-modes

The B-modes, in contrast to the E-modes, are generated by the tensor perturbations of the CMB, which are the primordial gravitational waves generated during the inflationary epoch.

As mentioned before in Subsection 2.1, the scalar perturbations do not generate B-mode polarization, so

$$C_{BB} = 0. \quad (17)$$

Again, the vector perturbations decay rapidly in the early Universe, so their contribution to the B-mode polarization is negligible compared to the tensor perturbations, which are the main source of B-mode polarization in the CMBR.

This is a great advantage for the detection of gravitational waves, as the B-mode polarization generated by the tensor perturbations is not contaminated by the scalar perturbations. However, there are other contaminants such as foreground emission from our galaxy, as always, and the lensing of the E-modes, as these are reshuffled or distorted, then a mixed E- and B-mode pattern is created (modulation).

The B-mode is expected to have about a tenth of the power of the E-mode, so it is much more challenging to detect. In order to compare the B-mode power spectrum with the E-mode, the temperature anisotropies (top panel) and the temperature-polarization cross-correlation (middle panel) power spectra, Fig. 7 is introduced.

The temperature anisotropy and the E-mode power spectrum have been well measured by many experiments, as can be seen in Fig. 7, and it is in good agreement with the predictions of the Λ CDM model. However, not enough experiments have been able to measure the B-mode power spectrum yet, although the little data obtained seems to be in agreement with the predictions.

The Planck experiment demonstrates that the correct model must generate perturbations that are outside the horizon at the time of recombination in order to explain the nonzero $C_{TE}(\ell)$ on large scales ($\ell \lesssim 100$). This fact is in agreement with the inflationary theory as it predicts that the initial perturbations were generated before recombination, during the inflationary epoch. There is no cross-correlation between temperature and magnetic polarization ($C_{TB} = 0$) nor between electric and magnetic polarization ($C_{EB} = 0$), as they are generated by different types of perturbations with different parity properties.

The lensing deflection power spectrum is also shown in Fig. 7 (bottom panel), as it is one of the main sources of contamination for the B-mode polarization. It is crucial to subtract the lensed E-mode contribution to the B-mode power spectrum in order to detect the primordial B-modes generated by the tensor perturbations.

It can be appreciated in Fig. 7 (all panels) that there is a general tendency for all power spectra to decay at large ℓ (small scales) because of the diffusion damping, which is caused by the finite thickness of the last scattering surface.

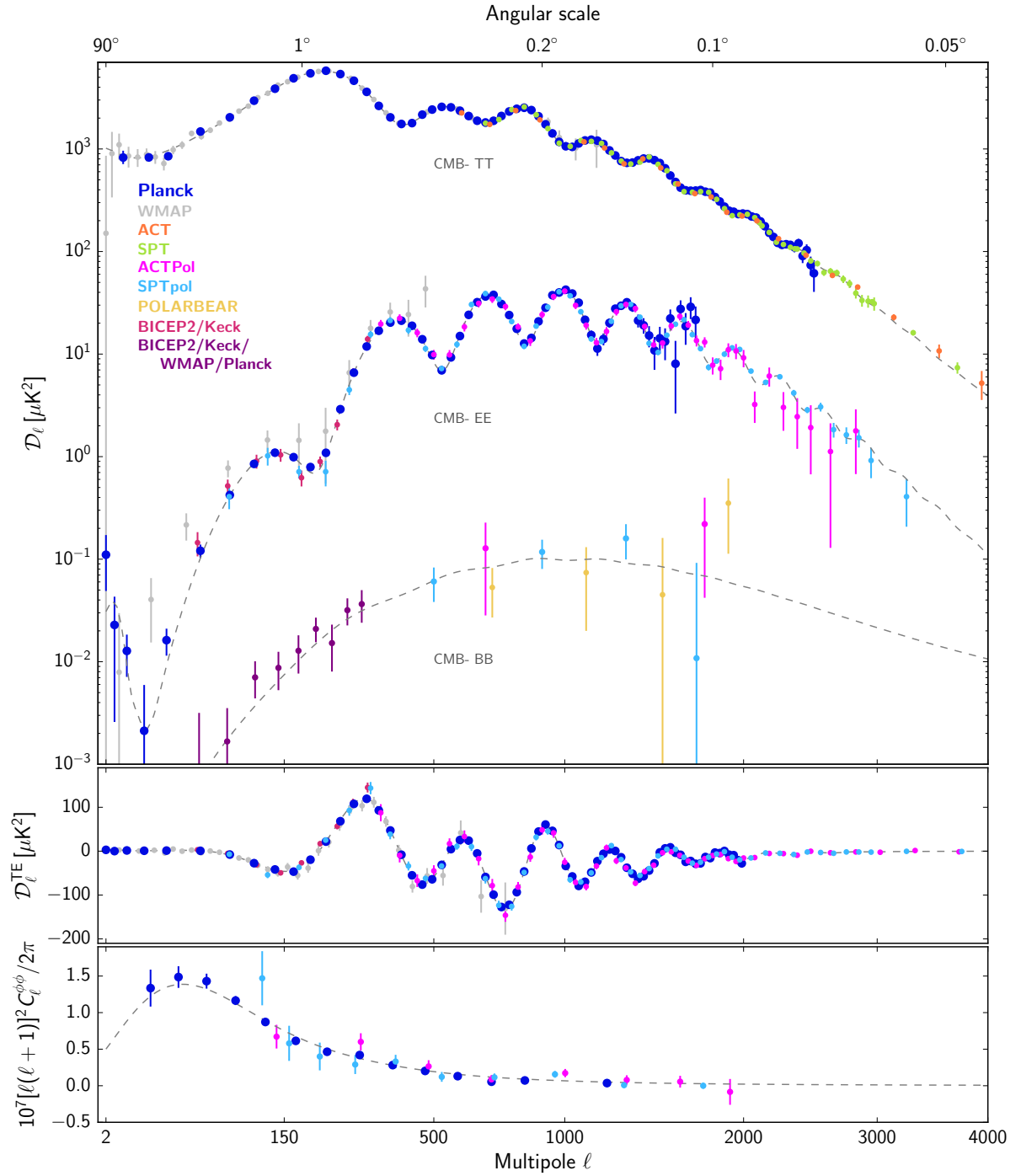


Fig. 7. Temperature, E-mode polarization, and B-mode polarization power spectra for the scalar, vector and tensor perturbations as a function of the multipole moment ℓ (top panel). Temperature-polarization cross-correlation power spectrum (middle panel). Lensing deflection power spectrum (bottom panel). Data from different experiments listed in the legend are shown as dots, while the predictions for the fiducial concordance cosmology are shown as dashed lines. Figure and footnote adapted from [3].

5 Current Observations and Future Prospects

There have been many experiments dedicated to study the CMB itself, the temperature anisotropies and the polarization. These have discovered the most important features of the CMB, such as the blackbody spectrum, the temperature anisotropies and the E-mode polarization, and they have set the current constraints on the cosmological parameters and the inflationary models.

I will list and comment on the most relevant ones in chronological order.

- **COBE:** Satellite launched in 1989 and operative until 1993 by NASA. It was the first experiment to determine that the CMBR is close to a blackbody spectrum and also to detect the temperature anisotropies of the CMB [5].
- **BOOMERanG:** Balloon-borne experiment launched in 1997 by a collaboration of Italian and American scientists. It was the first experiment to make high resolution measurements of the CMB temperature anisotropies, and it provided strong evidence for a flat Universe [6].
- **DASI:** Ground-based experiment located at the South Pole, operative from 2000 to 2008. It was the first experiment to detect the 2nd and 3rd acoustic peaks in the CMB temperature power spectrum. In 2002 detected the E-mode polarization of the CMB for the first time. It also measured the temperature-polarization cross-correlation power spectrum [7].
- **WMAP:** Satellite launched in 2001 and operative until 2010 by NASA. It's main mission was to measure the temperature anisotropies of the CMB with higher precision than the COBE satellite. It allowed to constrain the cosmological parameters with better accuracy such as the Hubble constant at 69.32 ± 0.80 km/s/Mpc, and to set the age of the Universe to about 13.8 billion years [8].
- **BICEP/Keck Array:** Series of ground-based cryogenic experiments located at the South Pole, operative from 2006 to present. They intend to measure the B-mode polarization of the CMB. In 2014, BICEP2 detected a pattern of B-mode polarization, although it was later shown that it was a contamination from the galactic dust [9].
- **Planck:** Satellite launched in 2009 and operative until 2013 by ESA. It improved the sensitivity and resolution of the WMAP satellite, so it provided a more detailed map of the CMB temperature anisotropies and polarization [10].
- **POLARBEAR:** Ground-based experiment located in the Atacama Desert operative from 2012 to present. It discovered the B-mode polarization generated by the lensing of the E-modes in 2014. It explores a deeper field than BICEP2, then the dust contamination has a smaller effect [11].
- **Simons Observatory:** Latest ground-based experiment located in the Atacama Desert. It is intended to improve the measurement of the tensor-to-scalar ratio r by an order of magnitude with respect to Planck satellite [12].

I will also discuss about the future experiments that pretend to improve the measurements of the CMB polarization, especially the B-modes, by having better sensitivity and resolution. If one of them is capable of detecting the primordial B-modes, it would be a smoking gun for inflation as it would provide direct evidence of the existence of primordial gravitational waves.

I will list and comment on the most promising future experiments.

- **LiteBIRD:** Satellite planned by JAXA for 2032, with the main mission of measuring the B-mode polarization of the CMB at different frequencies in order to separate the primordial B-modes from the foreground contamination [13].

- **PICO:** Satellite proposed by NASA with no launch date. It will consist of 21 frequency channels and a large number of detectors, so it will have a very high sensitivity to the CMB polarization [14].
- **PIPER:** Balloon-borne experiment planned by NASA with no launch date. It is meant to be capable of measuring the B-mode polarization three times smaller than the lowest predicted value [15].

6 Conclusion

I will list the general conclusions drawn from my investigation about the CMB polarization, and its decomposition into the E- and B-mode components, which are generated by the scalar, vector and tensor perturbations of the CMBR by Thomson scattering.

1. Electrons Thomson scattered by incoming photons with a quadrupole anisotropy generate linear polarization in the CMBR. The polarization can be decomposed into E- and B-modes, which have different parity and are generated by different types of perturbations.
2. The scalar (energy) density perturbations generate temperature anisotropies and E-mode polarization, but not B-mode polarization. In consequence, they first two are correlated, so there is a nonzero temperature-polarization cross-correlation power spectrum, $C_{TE}(\ell)$.
3. On the contrary, the tensor perturbations (primordial gravitational waves) generate both E- and B-mode polarization with similar amplitude, but do not generate temperature anisotropies, so there is no temperature-polarization cross-correlation power spectrum, $C_{TE}(\ell) = C_{TB}(\ell) = 0$ in this case.
4. The vector perturbations decay rapidly in the early Universe, so their contribution to the CMB polarization is negligible compared to the scalar and tensor perturbations.
5. The E-mode polarization has been well measured by many experiments, and it is in good agreement with the predictions of the Λ CDM model. However, not enough experiments have been able to properly measure the B-mode power spectrum yet, although the little data obtained seems to be in agreement with the predictions.
6. The detection of the B-mode polarization generated by the tensor perturbations would be a confirmation of the inflation theory, as it would provide direct evidence of the existence of primordial gravitational waves, which were generated during the inflationary epoch and that left a small imprint in the CMB polarization.
7. The future experiments that are planned to be launched in the next decade will hopefully be able to detect the primordial B-mode polarization and shed light on the physics of the early Universe, one of the most exciting and fundamental questions in cosmology.

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A Perturbations and Spherical Harmonics

In this appendix, we provide additional visualizations of the polarization patterns discussed in the main text.

A.1 Scalar Perturbations

The scalar perturbations of the CMB are represented by the spherical harmonic Y_2^0 . In the Fig. 8 the polarization pattern generated by a scalar perturbation in the x -direction is shown.

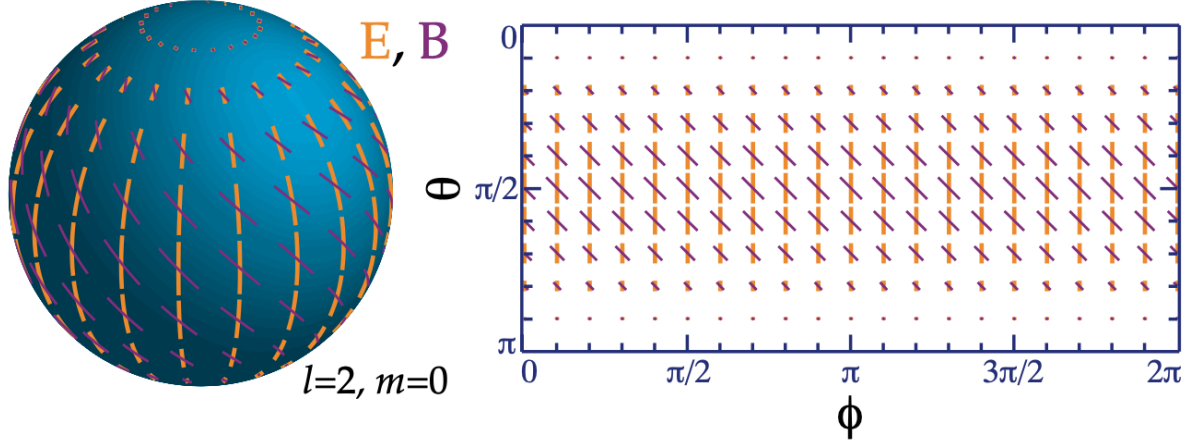


Fig. 8. Polarization pattern for the scalar perturbation with $(l, m) = (2, 0)$. The E-mode pattern (yellow, thick lines) is generated by the scattering of a scalar quadrupole perturbation, which has a quadrupole pattern with two peaks and two troughs along the z -axis. The B-mode pattern (purple, thin lines) is generated by the rotation of the E-mode pattern by 45° , which represents the orthogonal pattern with odd parity. FIG. 4 and footnote adapted from [2].

A.2 Vector Perturbations

The vector perturbations of the CMB are represented by the spherical harmonic $Y_2^{\pm 1}$. In the Fig. 9 the polarization pattern generated by a vector perturbation in the x -direction is shown.

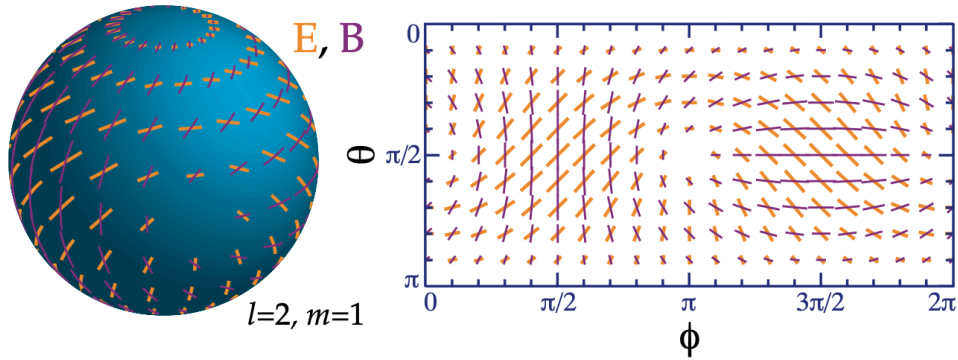


Fig. 9. Polarization pattern for the vector perturbation with $(l, m) = (2, 1)$. The E-mode pattern (yellow, thick lines) is generated by the scattering of a vector quadrupole perturbation, which has a quadrupole pattern with two peaks and two troughs along the z -axis. The B-mode pattern (purple, thin lines) is generated by the rotation of the E-mode pattern by 45° , which represents the orthogonal pattern with odd parity. FIG. 6 and footnote adapted from [2].

A.3 Tensor Perturbations

The tensor perturbations of the CMB are represented by the spherical harmonic $Y_2^{\pm 2}$. In the Fig. 10 the polarization pattern generated by a tensor perturbation in the x -direction is shown.

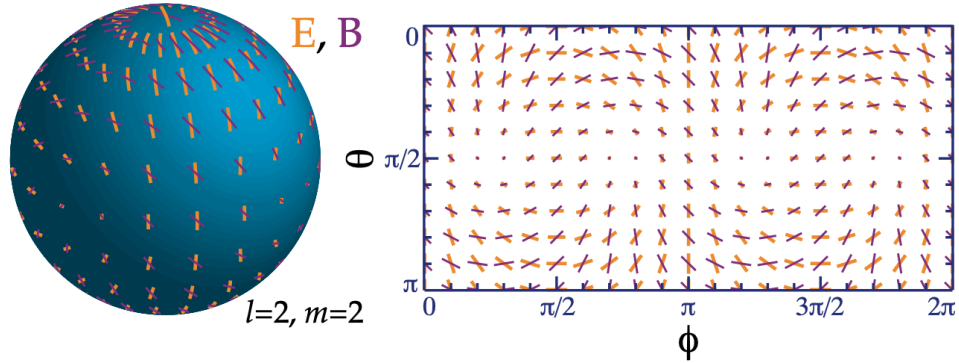


Fig. 10. Polarization pattern for the tensor perturbation with $(l, m) = (2, 2)$. The E-mode pattern (yellow, thick lines) is generated by the scattering of a tensor quadrupole perturbation, which has a quadrupole pattern with two peaks and two troughs along the z -axis. The B-mode pattern (purple, thin lines) is generated by the rotation of the E-mode pattern by 45° , which represents the orthogonal pattern with odd parity. FIG. 8 and footnote adapted from [2].